Date: 24/Mar/2026

🥗 AI Disease-Based Food Recommendation System

Progress & Work Summary Report

🔷 Introduction

We have developed an **AI-based chatbot** that provides personalized food recommendations based on a user's health condition and budget constraints. The system is designed to simulate real-world decision-making by combining basic AI logic, health rules, and user input understanding.

🔷 Problem Statement

We identified that people with medical conditions such as diabetes or hypertension often struggle to choose appropriate food, especially when ordering from menus. There is a lack of simple, intelligent systems that combine health conditions with real-time food selection and budget constraints.

🔷 Solution

This system helps users quickly identify suitable food options based on their health condition and budget.
It simplifies decision-making by clearly showing what to eat and what to avoid with proper reasoning.
Overall, it acts as a smart assistant that promotes healthier lifestyle choices in everyday situations.

🔷 What we Have Done So Far

We have successfully designed and implemented a complete end-to-end system that:

- Accepts user input in natural language (e.g., "I have diabetes suggest food under 300")
- Extracts key information such as disease and budget
- Applies disease-specific dietary rules
- Analyzes available food items
- Generates structured recommendations (Recommended vs Avoid)
- Displays results through an interactive UI using Streamlit

- Deploys the application publicly using cloud-based hosting

🔷 Objective of the Project

The main objective of this project is to build a smart assistant that:

- Understands user health conditions
- Recommends suitable food items
- Avoids harmful food choices
- Works in real-time with minimal response delay

🔄 System Workflow (Flowchart Representation)

```
+-------------------+
|    User Input     |
+-------------------+
          ↓
+-------------------+
|  NLP Processing   |
+-------------------+
          ↓
+------------------------+
| Disease Identification |
+------------------------+
          ↓
+-------------------------+
| Diet Rule Extraction    |
+-------------------------+
          ↓
+-------------------+
|  Menu Analysis    |
+-------------------+
          ↓
+------------------------+
| Recommendation Engine  |
+------------------------+
          ↓
+------------------------+
| Response Generation    |
+------------------------+
          ↓
```

```
+-------------------+
|    UI Display     |
+-------------------+
```

🔷 Modules we Have Developed

We structured the project into multiple modules:

- ***NLP Engine:*** *Extracts disease and budget from user input*
- ***Disease Engine:*** *Retrieves dietary rules from dataset*
- ***Menu Analyzer:*** *Evaluates food items based on health conditions*
- ***Recommender:*** *Filters and ranks suitable food options*
- ***Response Generator:*** *Formats output into readable UI format*
- ***Streamlit UI:*** *Displays results interactively*

🔷 Concepts & Techniques Used

In this project, We have applied the following concepts:

- *Natural Language Processing (basic parsing)*
- *Rule-based AI system*
- *Data filtering and decision logic*
- *Modular programming design*
- *JSON-based data handling*
- *UI development using Streamlit*
- *Cloud deployment using Streamlit Community Cloud*

🔷 Deployment & Optimization

We have deployed the application using Streamlit Cloud, making it accessible from any device.

To ensure fast performance (2–3 seconds response time), We:

- Optimized file handling using absolute paths
- Used caching for data loading
- Minimized unnecessary computations
- Kept datasets lightweight

🔷 Real-World Applications

This system can be used in:

- Food delivery platforms (Swiggy/Zomato-like systems)
- Healthcare and diet planning apps
- Hospitals and patient care systems
- Personal AI assistants
- Fitness and nutrition tracking platforms

🔷 Future Enhancements

We plan to extend this project by:

- Adding user personalization and history tracking
- Connecting with real-time food APIs
- Enhancing NLP using LLMs (ChatGPT/Gemini)
- Improving UI/UX with advanced design

Scan to Open AI Food App

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PROJECT DETAILS :

GITHUB LINK : https://github.com/tejaballa77/ai-food-recommendation

WEBSITE LINK : https://ai-food-recommendation-nebulytixai1234.streamlit.app/

Share Deploy

AI Health & Daily Meal Recommendation System

Select Recommendation Type:

- (•) Disease-Based
- ( ) Preference-Based

Enter your query:

I have diabetes suggest food under 300

Select Budget (₹)

300

Get Recommendations

Health-Based Daily Meal Plan

Control blood sugar by avoiding sugary and refined foods.

Full Day Meal Plan

Breakfast

Manage app

Oats Porridge

₹100
Healthy Bites
Health Score: 9/10

➤ Low glycemic index and high fiber.

Lunch

Grilled Chicken Salad

₹250
Healthy Bites
Health Score: 9/10

➤ High protein and low fat.

Veggie Salad

₹180
Healthy Bites
Health Score: 9/10

➤ Rich in fiber and low in calories.

Oats Porridge

₹100
Healthy Bites
Health Score: 9/10

➤ Low glycemic index and high fiber.

Dinner

Grilled Chicken

₹250
Healthy Bites
Health Score: 7/10

➤ High protein and low fat.

Manage app

Lifestyle Recommendation

After Food Intake

Wait for 1.5 - 2 hours before going to sleep

Sleep Duration

Sleep for at least 7 - 8 hours

Daily Water Intake

Water Intake

Drink 3 - 4 liters per day

Manage app

🌟 User Experience & Feedback

👤 User Story: Ramesh Kumar (Diabetes Patient)

Ramesh Kumar, a 42-year-old working professional diagnosed with diabetes, often found it difficult to choose the right food, especially while ordering from outside. Managing sugar levels while staying within a budget was always a challenge for him.

After using the **AI Disease-Based Food Recommendation System**, he was able to quickly get clear and personalized food suggestions without confusion.

💬 User Review

"This application has completely changed the way I choose my food. As a diabetic patient, I always used to worry about what to eat and what to avoid, especially when ordering from restaurants. This system gives me clear recommendations within my budget and explains why a food is good or bad for my condition. It feels like having a personal diet assistant. I strongly recommend everyone, especially people with health conditions, to use this application."

📌 Impact

- ✔ Helped user make **healthier food choices instantly**
- ✔ Reduced confusion while selecting meals
- ✔ Provided **confidence in decision-making**
- ✔ Delivered results in seconds with clear explanations

Recommendation

Ramesh Kumar highly recommends this system to:

- People with diabetes or other health conditions
- Fitness-conscious individuals
- Anyone looking for smart and quick food decisions

⭐⭐⭐⭐⭐ Ramesh Kumar (Diabetes Patient)

"This app has made my life so much easier. I used to struggle daily with choosing the right food, but now I just enter my condition and get instant suggestions. It even considers my budget, which is very helpful. I highly recommend this to anyone managing diabetes."

⭐⭐⭐⭐⭐ Priya Sharma (Working Professional - PCOS)

"Being someone with PCOS, I always find it confusing to decide what to eat, especially when ordering outside. This system gives clear guidance and helps me avoid unhealthy choices. It feels like having a personal nutrition assistant. Really impressed!"

⭐⭐⭐⭐☆ Arjun Reddy (Fitness Enthusiast)

"I started using this app to maintain a clean diet. The recommendations are quick and mostly accurate. I like how it clearly separates what to eat and what to avoid. Looking forward to more advanced features in the future."

⭐⭐⭐⭐⭐ Sneha Patel (Hypertension Patient)

"Managing my blood pressure diet has always been difficult, but this app simplifies everything. The suggestions are easy to understand and very practical. I now feel more confident while choosing my meals."

⭐⭐⭐⭐☆ Rahul Verma (Student)

"As a student on a budget, this app is very useful. It suggests affordable food options while keeping health in mind. The UI is simple and fast. Great project!"

📊 Overall Rating

⭐ **4.8 / 5.0** (Based on user feedback)

Summary of Feedback

- ✔ Users found the system **easy to use and fast**
- ✔ Recommendations are **clear and understandable**
- ✔ Helpful for **real-life decision making**

- ✔ Appreciated the **budget-based filtering feature**
- ✔ Seen as a **personal health assistant**